

Observed baseline extract  
2985 contacts in 924 households

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graph TD; A[Observed baseline extract<br/>2985 contacts in 924 households] --> B[Recorded TST: 2725 contacts<br/>877 households; 359 positive<br/>TST diameter ≥10 mm]; A --> C[Missing TST: 260 contacts<br/>47 households wholly unobserved]; B --> D[Assumed workflow: random order within each household<br/>Earlier members' results assumed returned before ranking<br/>No observed screening or result-return timestamps]; D --> E[Household-grouped five-fold CV<br/>20 seeds; fold-specific preprocessing<br/>Same household always in one fold]; D --> F[Paired random-forest models<br/>29 baseline versus 29 + 4 features<br/>Target's own result excluded]; E --> G[Separate evaluations: raw ranking, nested calibration and charged budget<br/>Conditional intervals distinguished from full-pipeline refit intervals<br/>No observed clinical implementation]; F --> G;
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Recorded TST: 2725 contacts  
877 households; 359 positive  
TST diameter  $\geq 10$  mm

Missing TST: 260 contacts  
47 households wholly unobserved

Assumed workflow: random order within each household  
Earlier members' results assumed returned before ranking  
No observed screening or result-return timestamps

Household-grouped five-fold CV  
20 seeds; fold-specific preprocessing  
Same household always in one fold

Paired random-forest models  
29 baseline versus 29 + 4 features  
Target's own result excluded

Separate evaluations: raw ranking, nested calibration and charged budget  
Conditional intervals distinguished from full-pipeline refit intervals  
No observed clinical implementation